

Kyungyoon Choung

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EDUCATION

Carnegie Mellon University

Master of Science

Jan 2026 - May 2026

Integrated Master's/Bachelor's in Mechanical Engineering

Bachelor of Science

Aug 2021 - Dec 2025

Major in Mechanical Engineering with additional major in Biomedical Engineering

QPA: 3.49/4.00; Dean's list in Fall 2024, Spring 2025, Fall 2025

EXPERIENCE

Research Undergraduate/Graduate researcher in Moore Lab

May 2024 - June 2026

- Evaluated accuracy and limitations of OpenCap (mobile mocap) against Optitrack through comparative data analysis in MATLAB
- Designed and machined female fixture end with standardized dimensions used for Instron testing grips

PROJECTS

3D Design Projects

PosturBot — Wearable Cervical Posture Monitor — 2026 - present

- Built dual-IMU wearable (MPU-6050, ESP32C3) with complementary filter for real-time cervical posture detection and BLE haptic feedback system
- Developed full-stack Chrome dashboard with live angle visualization and CSV export; validated BLE communication and progressive haptic feedback across 4 severity states

24672 DIY Design and Fabrication Projects — 2026

- Applied end-to-end fabrication workflows across 6 projects, using laser cutting, heat-formed acrylic, silicone molding, resin casting, and iterative foam/clay mockups, to translate CAD designs into functional physical prototypes.

24775 Bioinspired Robot Design and Experimentation: Undulating Gliding Snake Robot — 2026

- Developed MATLAB/Simulink simulation modeling undulatory gliding dynamics to evaluate body-wave effects on motion and stability for a multi-segment bio-inspired snake robot.

42402 BME Capstone: CIMT Switch Device for Children with CP — 2024-2025

- Developed an unconstrained wearable therapeutic switch device using Hall effect with wireless communication to support non-dominant hand rehabilitation of children with CP.

Coding Projects

24703 Real-Time Posture Correction and Kinematic Alignment — 2026

- Developed a real-time yoga pose correction system in Python using Google MediaPipe and a custom Jacobi SVD algorithm to compare user poses against ground truth and provide joint-specific feedback.

15112 Term Project: Clean CMU — 2022

- Designed an educational game on proper trash disposal, integrating physics-based mechanics and the hyperbola formula for interactive gameplay in a maze.

ACTIVITIES

KPDC: K-Pop Dancing Club 2022- Present

- Led choreography instruction, formation strategy, and center/solo role allocation; managed scheduling, member coordination, and semester-end performance film.

SKILLS

Programming: Python, C++, Matlab

Technical experience: Simulink, OpenCap, WorkBench, Optitrack, Arduino, ANSYS fluent

Fabrication tools: Solidworks, Laser cutting, Fusion, 3D printing, CNC mill, Lathe

Languages: Fluent in Korean, Fluent in English